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Introduction to the dataset

Given Social_Network_Ads csv file has 400 rows and with 5 columns, which are User_ID, Gender, Age, EstimatedSalary and Purchased. Since User_ID does not give proper meaning, we drop the User_ID in rows.

Also Purchased column will be our target variable. Remaining Gender, Age and EstimatedSalary columns will be our input variable.

The target variable falls under the categorical data, which is purchased or not-purchased, so we analyze the data with classification algorithms instead of regression.

The dataset's value counts display as follows.
Total rows: 400
Not-Purchased: 257 (64.25%)
Purchased: 143 (35.75%)
These figures clearly show that the dataset is imbalanced.

Plan

This document will analysis the confusion matrix and classification report with the following classification algorithms.

1. DecisionTreeClassfier
2. RanfomForestClassfier
3. SVM Classifier
4. Logistic Regression

DecisionTreeClassfier

Please refer to Classification-Algorithms\Classification_DT.ipynb notebook in the repo.

		ACTUAL	
		Purchased	Not Purchased
PREDICTED	Purchased	71	8
	Not Purchased	3	38

Confusion matrix for DecisionTreeClassifier display as above. Since True-Purchased and True-NotPurchased are showing good values than False-Purchased and False-NotPurchased, but note that the TYPE-I error is slightly higher than the TYPE-II error, which is (8 > 3). So cannot be considered as a good model.

```
              precision    recall  f1-score   support

           0       0.96      0.90      0.93        79
           1       0.83      0.93      0.87        41

    accuracy                           0.91       120
   macro avg       0.89      0.91      0.90       120
weighted avg       0.91      0.91      0.91       120
```

Accuracy is 0.91 and precision/recall values also seems good but the TYPE-I and TYPE-II values are in the reverse direction, we might go with another model.

RanfomForestClassfier

Please refer to Classification-Algorithms\Classification_RF.ipynb notebook in the repo.

ACTUAL

PREDICTED	Purchased	Not Purchased
Purchased	73	6
Not Purchased	5	36

Above figures show the confusion matrix of RandomForesetClassifier. Same like DecisionTreeClassifier both True-Purchased and True-NotPurchased have good values and the same time the TYPE-I error is slightly higher than TYPE-II error (since this is not good sign, better to consider other algorithms for this dataset)

```
              precision    recall  f1-score   support

           0       0.94      0.92      0.93        79
           1       0.86      0.88      0.87        41

    accuracy                           0.91       120
   macro avg       0.90      0.90      0.90       120
weighted avg       0.91      0.91      0.91       120
```

Same like DecisionTreeClassifire, the accuracy is 0.91 and precision/recall values also good but the TYPE-I and TYPE-II values are in the reverse direction, we might go with another model.

SVM Classifier

Please refer to Classification-Algorithms\Classification_SVM.ipynb notebook in the repo.

ACTUAL

PREDICTED	Purchased	Not Purchased
Purchased	77	2
Not Purchased	23	18

This is for SVM Classfier's confusion matrix. Event though the TYPE-I error is better than TYPE-II error (2 < 23), the True-NotPurchased value is worse than Fasle-NotPurchased value. So cannot consider as good model.

	precision	recall	f1-score	support
0	0.77	0.97	0.86	79
1	0.90	0.44	0.59	41
accuracy			0.79	120
macro avg	0.83	0.71	0.73	120
weighted avg	0.81	0.79	0.77	120

SVM's accuracy is poor among 4 models and recall value for purchased category is low. Precision and re-call figures must be high for a good model.

Since TYPE-I and TYPE-II errors also in the opposite direction and the classification report also not satisfied, we can reject this mode

Logistic Regression

Please refer to Classification-Algorithms\Classification_Logistic.ipynb notebook in the repo.

ACTUAL

PREDICTED	Purchased	Not Purchased
Purchased	74	5
Not Purchased	8	33

Based on confusion matrix Logistic Regression shows good figures. Also the TYPE-1 is better than the TYPE-II (5 < 8).

```
              precision    recall  f1-score   support

           0       0.90      0.94      0.92        79
           1       0.87      0.80      0.84        41

    accuracy                           0.89       120
   macro avg       0.89      0.87      0.88       120
weighted avg       0.89      0.89      0.89       120
```

Even through the accuracy bit low than DecisionTree and RandomForest, its giving the good precision/recall values for Not-Purchased category and bit lower values for Purchased category. It's completely acceptable, because the dataset is imbalanced as mentioned in the introduction section.

Among the 4 models, Logistic Regression model only gives the good TYPE-1 and TYPE-II values.

Final Note:

Confusion Matrix	
DT	True values good, but Type-I > Type-II
RF	True values good, but Type-I > Type-II
SVM	Type-1 < Type-II error, but the False-NotPuschased is higher than True-NotPurchased.
Logistic	True values good also Type-I < Type-II

		CLASSIFICATIOIN REPORT				MATRIX ANALYSIS
ALGORITHM		ACCURACY	PRECISION	RECALL	F1-SCORE	
DecisionTreeClassfier	Not-Purchased	0.91	0.96	0.90	0.93	TYPE 1 > TYPE 2
	Purchased		0.83	0.93	0.87	
RanfomForestClassfier	Not-Purchased	0.91	0.94	0.92	0.93	TYPE 1 > TYPE 2
	Purchased		0.86	0.88	0.87	
SVM Classifier	Not-Purchased	0.79	0.77	0.97	0.86	TYPE 1 > TYPE 2
	Purchased		0.90	0.44	0.59	
Logistic Regression	Not-Purchased	0.89	0.90	0.94	0.92	TYPE 1 < TYPE 2
	Purchased		0.87	0.80	0.84	

Logistic Algorithm wins here.